

SARANSH SINGH

Security Research · Offensive Tooling · Detection Engineering

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EDUCATION

B.Tech in Electronics & Communication Engineering

Deenbandhu Chhotu Ram University of Science & Technology, Murthal

2023 – 2027

EXPERIENCE

Cyber Security Intern | SafeSkillz Ltd, London (Remote)

May 2026 – Jun 2026

- Completed Cybersecurity Internship Programme (16 May – 28 Jun 2026) with instructor-led sessions, weekly assessments, and practical labs in network security, threat intelligence, web application security, and DFIR.
- Android mobile application pentesting and system exploitation using Metasploit Framework, covering Linux and Windows privilege escalation.
- Capstone Project – Vulnerable Machine Assessment – with technical report documenting methodology, findings, and remediation. Applied UK GDPR compliance.

Security Researcher | Independent / CTF Teams

2024 – Present

- CTF competitor in web exploitation, binary exploitation, cryptography, and forensics. Built custom tooling for automated recon, vulnerability scanning, and exploitation workflows.
- Published technical writeups on C2 architecture, vulnerability research methodology, and detection engineering.

SOC / Detection Engineering (Lab) | Home Lab

2024 – Present

- Deployed Wazuh SIEM with 40+ MITRE ATT&CK-mapped detection rules (privilege escalation, persistence, credential access, lateral movement).
- Automated alert pipeline with Python dedup, MITRE enrichment, and Slack routing. Simulated TTPs (Kerberoasting, injection, DCSync) to validate detection coverage.

Offensive Security Training | HTB, TryHackMe, Proving Grounds

2023 – 2024

- 50+ AD, web, and network pentesting lab machines with structured methodology covering initial access through domain dominance.

PROJECTS

ShadowGate – C2 Framework

- HTTP/S C2 with AES-256-GCM encrypted beaconing, modular payload plugins, jittered intervals, server-side task queuing with per-implant key negotiation. OpSec features include user-agent rotation and URL obfuscation.

Wazuh Sentinel – SIEM Detection Engine

- 40+ MITRE ATT&CK-mapped detection rules on Docker-based Wazuh; automated alert pipeline with dedup, enrichment, and severity-based Slack notification routing.

BinaryForge – Exploitation Framework

- ROP chain builder, buffer overflow generator, format string tester for x86/x64 Linux. Implements bypass techniques for NX, stack canaries, and PIE with documented workflow.

PacketSentry – Network IDS

- Real-time NIDS using raw sockets and BPF on Linux. Signature engine detects port scans, ARP spoofing, DNS tunneling, and SYN floods with live alerting.

ForgeProxy – Intercept Proxy

- HTTP/HTTPS intercept proxy with request/response inspection, CA cert generation for TLS interception, WebSocket support, and automated fuzzing integration.

SKILLS

Offensive	Binary exploitation, web app pentesting, AD attacks, Metasploit, C2 development, adversary emulation
Defensive	Wazuh SIEM, detection engineering, MITRE ATT&CK, log analysis, incident response, alert automation
Languages	Python, C, Bash, PowerShell, SQL, Java, JavaScript
Tools	Burp Suite, Wireshark, Ghidra, BloodHound, Impacket, Hashcat, Docker, Git, VirtualBox